

CBSE Test Paper 03
Chapter 3 Motion in A Straight Line

1. If the position- time graph is a straight line 1

- a. The velocity is increasing
- b. The velocity is decreasing
- c. The velocity is zero
- d. The velocity is constant

2. Two parallel rail tracks run north-south. Train A moves north with a speed of 54 km/hr, and train B moves south with a speed of 90 km/hr. What is the velocity of ground with respect to B in m/s? Choose the positive direction of x-axis to be from south to north. 1

- a. 22
- b. 30
- c. 25.0
- d. 28

3. A car moving at 60 km/h comes to a stop in 4.0 s. What was its deceleration? 1

- a. 20 m/s^2
- b. -30 m/s^2
- c. -15 m/s^2
- d. -25 m/s^2

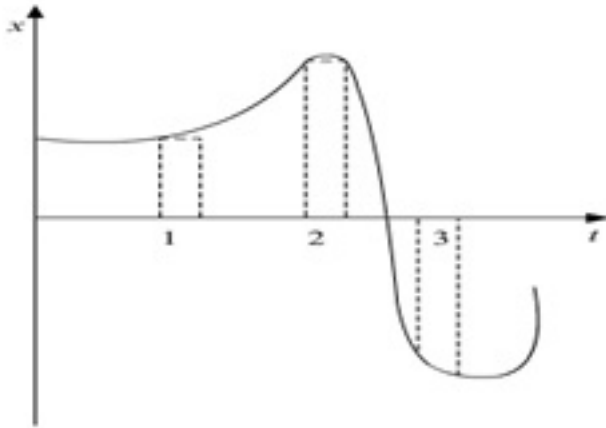
4. A ball starts from rest and accelerates at 0.500 m/s^2 while moving down an inclined plane 9.00 m long. When it reaches the bottom, the ball rolls up another plane, where, after moving 15.0 m, it comes to rest. How many seconds does it take to roll down the first plane? 1

- a. 8.00
 - b. 4.00
 - c. 6.0
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d. 2.00

5. A truck has a velocity of 3 m /s at time $t=0$. It accelerates at 3 m / s^2 on seeing police .What is its velocity in m/s at a time of 2 sec **1**
- a. 7
b. 9
c. 12
d. 8
6. The v-t graphs of two objects make angle 30° and 60° with the time axis. Find the ratio of their accelerations. **1**
7. Two balls of different masses are thrown vertically upward with same initial speed. Which one will rise to a greater height? **1**
8. Suppose two trains A and B are moving with uniform velocities along parallel tracks in the same direction and the velocities of A and B be 60 km /h in East and 65 km/h in East. Find the relative velocity of B w.r.t. A. **1**
9. The displacement of a body is proportional to t^3 , where t is time elapsed. What is the nature of acceleration-time graph of the body? **2**
10. The odometer of Raja's car reads 1700 km at the start of a trip and 2500 km at the end of the trip. The trip took 16 h. What is the average speed of Raja's car in ms^{-1} ? **2**
11. The position of an object is given by $x = 2t^2 + 3t$. Find out that its motion is uniform or non - uniform. **2**
12. Two town A and B are connected by a regular bus service with a bus leaving in either direction every T min. A man cycling with a speed of 20 kmh^{-1} in the direction A to B notices that a bus goes past him every 18 min in the direction of his motion, and every 6 min in the opposite direction. What is the period T of the bus service and with what speed do the buses ply of the road? **3**
13. Is earth inertial or non - inertial frame of reference? **3**
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14. Below figure gives the x - t plot of a particle in one-dimensional motion. Three different equal intervals of time are shown. In which interval is the average speed greatest, and in which is it the least? Give the sign of average velocity for each interval. **3**



15. A ball is thrown upward with an initial velocity of 100 m/s. After how much time will it return? Draw velocity-time graph for the ball and find from the graph. **5**
- Maximum height attained by the ball.
 - Height of the ball after 15 s. Take $g = 10 \text{ ms}^{-1}$.